In[118]:=

```
ClearAll["Global`*"];
SeedRandom[777];

(* 1. 超高速大视界并发引擎 (Window = 200) *)
StepEvolution[g_Graph, window_] := Module[
 {allEdges, activeEdges, nActive, used, updates,
  tempE1, tempE2, e1List, x, y, z, wCounter, newVertices,
  newActive, newFrozen, remainingEdges},

 allEdges = EdgeList[g];
 activeEdges = RandomSample[Cases[allEdges, _UndirectedEdge]];
 nActive = Length[activeEdges];
 If[nActive < 2, Return[g]];

 used = ConstantArray[False, nActive];
 updates = {};

 Do[
  If[! used[[i]],
   tempE1 = activeEdges[[i]];
   e1List = List @@ tempE1;
   Do[
    If[! used[[j]],
     tempE2 = activeEdges[[j]];
     If[Length[Intersection[e1List, List @@ tempE2]] == 1,
      AppendTo[updates, {tempE1, tempE2}];
      used[[i]] = True;
      used[[j]] = True;
      Break[];
     ]
    ],
    {j, i + 1, Min[i + window, nActive]}
   ]
  ],
  {i, 1, nActive}
 ];

 If[Length[updates] == 0, Return[g]];
 wCounter = Max[VertexList[g]];
 newVertices = {}; newActive = {}; newFrozen = {};
```

```
 Do[
  y = Intersection[List @@ pair[[1]], List @@ pair[[2]]][[1]];
  x = Complement[List @@ pair[[1]], {y}][[1]];
  z = Complement[List @@ pair[[2]], {y}][[1]];
  wCounter = wCounter + 1;
  AppendTo[newVertices, wCounter];
  newActive = Join[newActive,
       {UndirectedEdge[x, z], UndirectedEdge[x, wCounter], UndirectedEdge[wCounter, z]}];
  newFrozen = Join[newFrozen, {DirectedEdge[x, y], DirectedEdge[y, z]}];,
  {pair, updates}
 ];

 remainingEdges = Complement[allEdges, Flatten[updates]];
 Graph[VertexList[g] ~Join~ newVertices, Union[remainingEdges, newActive, newFrozen]]
];

(* 2. 初始化与背景演化 *)
vA = Range[1, 8]; eA = EdgeList[CompleteGraph[8]];
vB = Range[9, 14]; eB = EdgeList[CompleteGraph[6]] /. (i_Integer :> i + 8);
bridge = {UndirectedEdge[8, 9]};
initG = Graph[Join[vA, vB], Join[eA, eB, bridge]];

steps = 40;
evolutionHistory = {initG};
Print[Style["正在生成高相干宇宙背景 (Window=200)...", Orange, Bold]];
Monitor[
 Do[AppendTo[evolutionHistory,
    StepEvolution[Last[evolutionHistory], 100]], {i, 1, steps}],
 Row[{"已演化步数: ", i, "/", steps}]
];

(* 3. 【核心修正】提取全息骨架与活性波前 *)
finalG = Last[evolutionHistory];

(* 提取表皮：只用来观测和寻找波源 *)
activeEdges = Cases[EdgeList[finalG], _UndirectedEdge];
activeG = Graph[activeEdges];
activeNodes = VertexList[activeG];
Print["当前活性表皮节点数: ", Length[activeNodes]];
```

```
(* 提取骨架：把全图（包含冻结边）转为无向流形，作为波传播的真实介质 *)
bulkG = Graph[VertexList[finalG], UndirectedEdge @@@ EdgeList[finalG]];

(* 4. 寻找“双缝”源 (S1 和 S2) *)
Print[];
s1 = RandomChoice[activeNodes];
(* 注意：现在的距离条件是基于全息骨架 bulkG 寻找的 *)
s2Candidates = Select[activeNodes, GraphDistance[bulkG, s1, #] == 6 &];
s2 = If[Length[s2Candidates] > 0, First[s2Candidates], RandomChoice[activeNodes]];

(* 5. 穿透骨架的干涉计算 *)
(* 信号通过宇宙的底层骨架进行传播测距，彻底避免表皮断裂带来的 Infinity *)
d1 = N[GraphDistance[bulkG, s1, #]] & /@ activeNodes;
d2 = N[GraphDistance[bulkG, s2, #]] & /@ activeNodes;

(* ===== 替换原来的第 6 步和第 7 步 ===== *)

(* 微调波长为无理数，打破相位的整数锁死，涌现连续光谱 *)
lambda = 20.0;
intensity = (Cos[(2.0 * Pi * d1) / lambda] + Cos[(2.0 * Pi * d2) / lambda])^2;

Print[Style["正在渲染【全息骨架】上的宏观干涉条纹...", Blue, Bold]];
colors =
  Blend[{{0, Black}, {1, Darker[Blue]}, {2, Cyan}, {3, Yellow}, {4, White}}, #/4] & /@
   intensity;

(* 核心：渲染完整的 bulkG，而不是破碎的表皮，提取最大连通图以防万一 *)
mainSpacetime = First[ConnectedGraphComponents[bulkG]];
plotNodes = VertexList[mainSpacetime];
plotColors = Thread[activeNodes → colors];
(* 将非活性节点（历史骨架）染成暗灰色，作为背景衬托 *)
finalColors =
  # → If[MemberQ[activeNodes, #], # /. plotColors, Darker[Gray, 0.8]] & /@ plotNodes;

plotInterference = GraphPlot[mainSpacetime,
 VertexStyle → finalColors,
 VertexSize → If[MemberQ[activeNodes, #], 0.8, 0.2] & /@ plotNodes,
 EdgeStyle → Directive[Opacity[0.15], LightGray],
 PlotLabel → Style["Macroscopic Double-Slit in Connected Spacetime", 14, Bold],
 Background → Black,
```

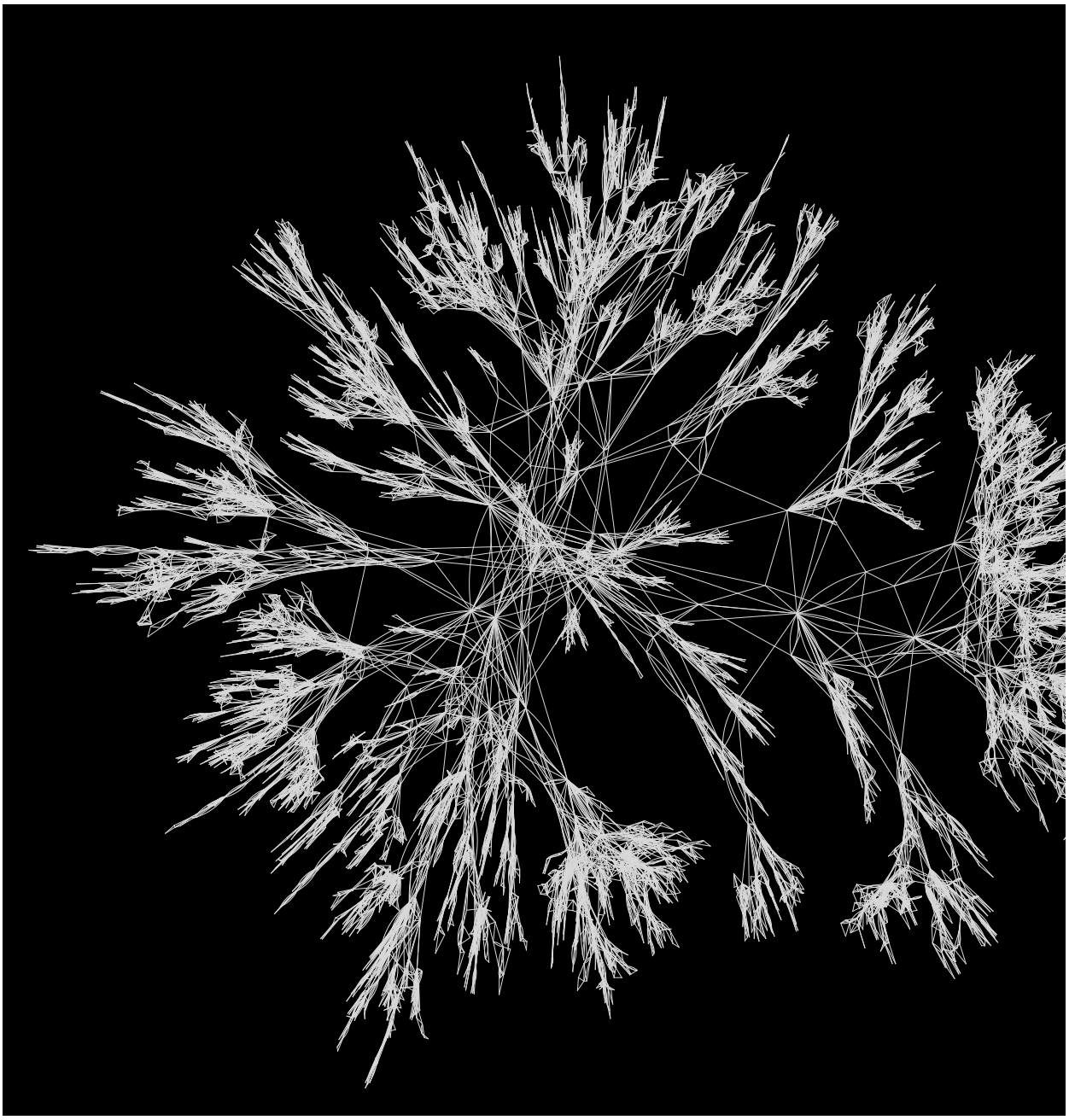
```
 ImageSize → 800
];

Print[plotInterference];

histogram = Histogram[intensity, 30, "Probability",
 Frame → True, FrameLabel → {"Interference Intensity", "Probability"},
 PlotLabel → "Continuous Intensity Distribution (Irrational Lambda)",
 ChartStyle → Cyan, ImageSize → 400
];
Print[histogram];
```

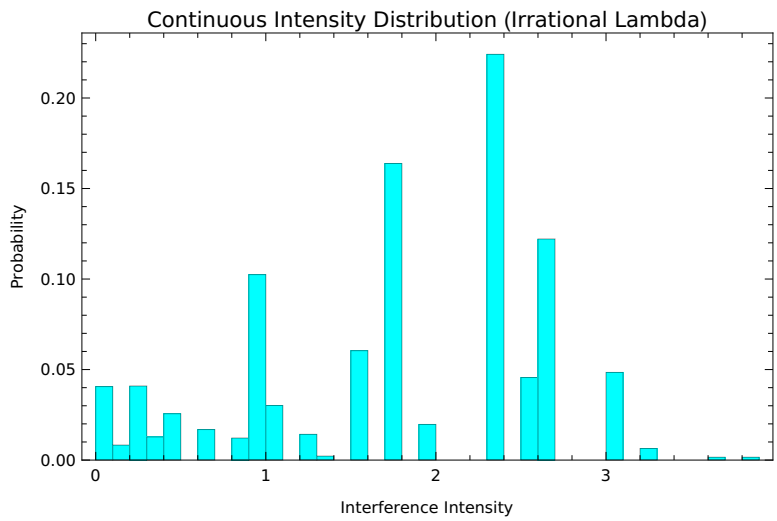
正在生成高相干宇宙背景 (Window=200)...

当前活性表皮节点数: 6403

正在定位拓扑双缝源...

正在渲染【全息骨架】上的宏观干涉条纹...





In[153]:=

```
ClearAll["Global`*"];
SeedRandom[777];

(* 1. 超高速大视界并发引擎 (Window = 200) *)
StepEvolution[g_Graph, window_] := Module[
 {allEdges, activeEdges, nActive, used, updates,
  tempE1, tempE2, e1List, x, y, z, wCounter, newVertices,
  newActive, newFrozen, remainingEdges},

 allEdges = EdgeList[g];
 activeEdges = RandomSample[Cases[allEdges, _UndirectedEdge]];
 nActive = Length[activeEdges];
 If[nActive < 2, Return[g]];

 used = ConstantArray[False, nActive];
 updates = {};

 Do[
  If[! used[[i]],
   tempE1 = activeEdges[[i]];
   e1List = List @@ tempE1;
   Do[
    If[! used[[j]],
     tempE2 = activeEdges[[j]];
     If[Length[Intersection[e1List, List @@ tempE2]] == 1,
      AppendTo[updates, {tempE1, tempE2}];
      used[[i]] = True;
      used[[j]] = True;
```

```
      Break[];
     ]
    ],
    {j, i + 1, Min[i + window, nActive]}
   ]
  ],
  {i, 1, nActive}
 ];

 If[Length[updates] == 0, Return[g]];
 wCounter = Max[VertexList[g]];
 newVertices = {}; newActive = {}; newFrozen = {};

 Do[
  y = Intersection[List @@ pair[[1]], List @@ pair[[2]]][[1]];
  x = Complement[List @@ pair[[1]], {y}][[1]];
  z = Complement[List @@ pair[[2]], {y}][[1]];
  wCounter = wCounter + 1;
  AppendTo[newVertices, wCounter];
  newActive = Join[newActive,
       {UndirectedEdge[x, z], UndirectedEdge[x, wCounter], UndirectedEdge[wCounter, z]}];
  newFrozen = Join[newFrozen, {DirectedEdge[x, y], DirectedEdge[y, z]}];,
  {pair, updates}
 ];

 remainingEdges = Complement[allEdges, Flatten[updates]];
 Graph[VertexList[g] ~Join~ newVertices, Union[remainingEdges, newActive, newFrozen]]
];

(* 2. 初始化与生成未观测的相干背景 (40步) *)
vA = Range[1, 8]; eA = EdgeList[CompleteGraph[8]];
vB = Range[9, 14]; eB = EdgeList[CompleteGraph[6]] /. (i_Integer :> i + 8);
bridge = {UndirectedEdge[8, 9]};
initG = Graph[Join[vA, vB], Join[eA, eB, bridge]];

steps = 40;
evolutionHistory = {initG};
Print[Style["1. 正在生成高相干宇宙背景 (纯净叠加态)...", Orange, Bold]];
Monitor[
```

```
 Do[AppendTo[evolutionHistory,
    StepEvolution[Last[evolutionHistory], 100]], {i, 1, steps}],
 Row[{"已演化步数: ", i, "/", steps}]
];

finalG = Last[evolutionHistory];
activeNodes = VertexList[Graph[Cases[EdgeList[finalG], _UndirectedEdge]]];
bulkG = Graph[VertexList[finalG], UndirectedEdge @@@ EdgeList[finalG]];

(* 3. 寻找“双缝”源 S1 和 S2 *)
Print[];
s1 = RandomChoice[activeNodes];
s2Candidates = Select[activeNodes, GraphDistance[bulkG, s1, #] == 6 &];
s2 = If[Length[s2Candidates] > 0, First[s2Candidates], RandomChoice[activeNodes]];

(* 4. 引入宏观观测者（波函数坍缩开始！） *)
Print[Style[, Red, Bold]];

obsSize = 50; (* 这里可以随便改你的探测器“拓扑质量”，比如 50, 100, 150 *)

observerNodes = Range[Max[VertexList[finalG]] + 1, Max[VertexList[finalG]] + obsSize];
(* 制造一个极其致密的活性网络作为探测器 *)
observerEdges = EdgeList[CompleteGraph[obsSize]] /. (i_Integer :> observerNodes[[i]]);
(* 将探测器与 S1 进行拓扑纠缠（连接） *)
measurementEdge = {UndirectedEdge[s1, observerNodes[[1]]]};

measuredG = Graph[Join[VertexList[finalG], observerNodes],
       Join[EdgeList[finalG], observerEdges, measurementEdge]];

(* 让宇宙带着观测者再演化 10 步，确保算力被彻底抽干 *)
stepsDecoherence = 10;
evolutionMeasured = {measuredG};
Monitor[
 Do[AppendTo[evolutionMeasured, StepEvolution[Last[evolutionMeasured], 100]],
   {i, 1, stepsDecoherence}],
 Row[{"测量互动进行中: ", i, "/", stepsDecoherence}]
];

(* 5. 提取坍缩后的全息骨架 *)
collapsedG = Last[evolutionMeasured];
```

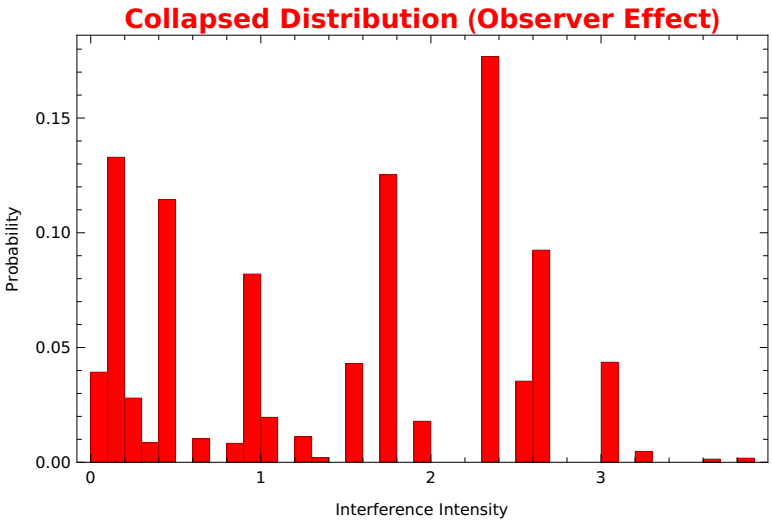
```
bulkCollapsed =
  Graph[VertexList[collapsedG], UndirectedEdge @@@ EdgeList[collapsedG]];
activeNodesCollapsed = VertexList[Graph[Cases[EdgeList[collapsedG], _UndirectedEdge]]];

(* 6. 重新计算测地线相位 *)
d1 = N[GraphDistance[bulkCollapsed, s1, #]] & /@ activeNodesCollapsed;
d2 = N[GraphDistance[bulkCollapsed, s2, #]] & /@ activeNodesCollapsed;

lambda = 20.0;
intensityCollapsed = (Cos[(2.0 * Pi * d1) / lambda] + Cos[(2.0 * Pi * d2) / lambda])^2;

(* 7. 终极对比：坍缩后的直方图 *)
Print[Style[, Red, Bold]];

histogramCollapsed = Histogram[intensityCollapsed, 30, "Probability",
 Frame → True, FrameLabel → {"Interference Intensity", "Probability"},
 PlotLabel → Style["Collapsed Distribution (Observer Effect)", 14, Bold, Red],
 ChartStyle → Red, ImageSize → 400
];
Print[histogramCollapsed];

Print[Style["\n[终极判决]", Blue, Bold, 16]];
Print["对比上一张带有多个波峰的干涉图，观察这张红色的直方图："];
```

1. 正在生成高相干宇宙背景 (纯净叠加态)...

正在定位拓扑双缝源...

2. 宏观探测器接入 S1，引发拓扑强退相干！

3. 渲染坍缩后的概率分布...



[终极判决]

对比上一张带有多个波峰的干涉图，观察这张红色的直方图：